

TOBB University of Economics and Technology

Department of Electrical and Electronics Engineering

ELE 224L – Electronic Circuits Laboratory

Experiment 2 – Preliminary Work: MOSFET Circuits

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1 INTRODUCTION

1.1 Objective of the Experiment

The objective of this experiment is to learn the operating principle of enhancement-mode MOSFETs, the DC biasing of a MOSFET with a resistive voltage divider, and the operation of the CMOS inverter, which is the basic cell of CMOS logic. The BS170 (N-channel) and BS250 (P-channel) transistors are studied through their datasheet parameters, hand calculations with the square-law model, and SPICE simulations.

1.2 Benefits of the Experiment

The experiment shows how the datasheet parameters (V_{TH}, $I_{D|max|}$, pin order) are obtained and used, how the three operating regions (cut-off, saturation, triode) are identified from the voltages, and why a complementary pair conducts significant current only while the input is switching. It also develops the skills of drawing a circuit on a breadboard without wiring errors and of explaining the differences between a simple analytical model and a SPICE model.

2 CALCULATIONS AND SIMULATIONS

2.1 MOSFET Properties

The parameters of the two transistors were taken from the datasheets and are listed in Table 2.1.1. The pin order of the two transistors is opposite: the source of the BS250 is on pin 1, whereas the source of the BS170 is on pin 3. Since both packages look identical and both are used in the same inverter circuit, this difference must be considered during wiring. The pin order of the BS250 was derived from the bottom-view drawing in its datasheet (D–G–S from below, i.e., S–G–D when the flat face is towards the viewer), and it should be confirmed on the physical part because the datasheet states that other pin configurations exist on special request. The two datasheets of the BS250 give slightly different limits (−1.0 to −3.0 V and −1 to −3.5 V for the threshold, 250 mA and 230 mA for the current) because they belong to different manufacturers, and the typical values are used in the calculations.

Table 2.1.1 MOSFET parameters taken from the datasheets and TO-92 pin order

Parameter	BS170 (NMOS)	BS250 (PMOS)
$V_{GS(th)}$ min / typ / max	0.8 / 2.0 / 3.0 V	-1.0 / -2.0 / -3.0 V (General Semiconductor); -1 / – / -3.5 V (Diodes Inc.)
V_{TH} used in calculations	2.0 V (typical)	-2.0 V (typical)
$I_{D(max)}$ (continuous)	500 mA	-250 mA (General Semiconductor); -230 mA (Diodes Inc.)
Pin 1 / Pin 2 / Pin 3 (flat face towards the viewer, leads down)	Drain / Gate / Source	Source / Gate / Drain

2.2 CMOS Inverter Circuit

The inverter was drawn on the breadboard as shown in Figure 2.2.1, and its circuit diagram is shown in Figure 2.2.2. The source pin (S) of the PMOS is connected to V_{DD}, the source pin (S) of the NMOS is connected to ground, the two gates are tied together to form V_{IN}, and the two drains are tied together to form V_{OUT}. The pin order of each transistor is observed in the drawing (S–G–D for the BS250 and D–G–S for the BS170). The connections in the drawing were checked automatically against the circuit diagram of the experiment sheet.

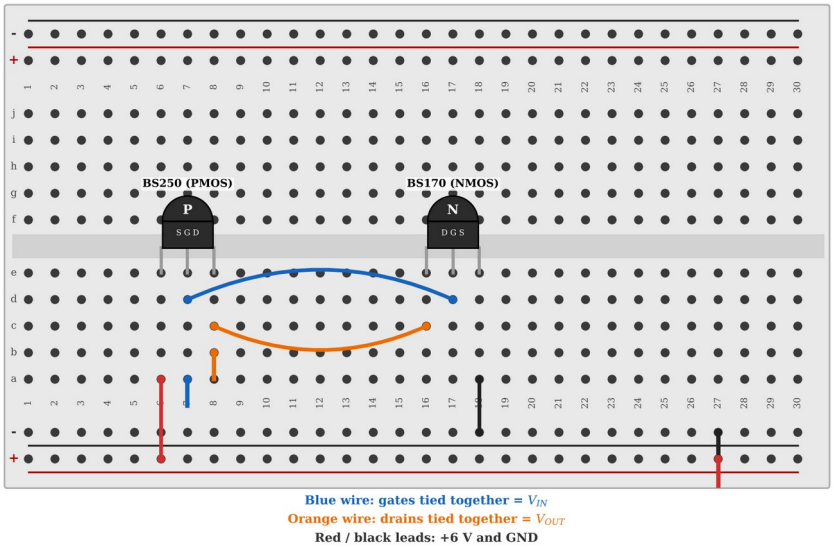


Figure 2.2.1 CMOS inverter on the breadboard (BS250 PMOS on columns 6–8, BS170 NMOS on columns 16–18).

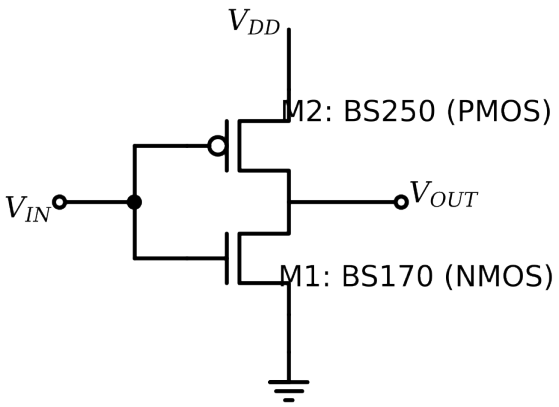


Figure 2.2.2 CMOS inverter circuit diagram.

Operating principle. When V_{IN} is below the NMOS threshold ($V_{TN}=2.1$ V in the given model), the NMOS is in cut-off and the PMOS (with $V_{SG}=V_{DD}-V_{IN}>|V_{TP}|$) pulls the output to $V_{DD}=6$ V. When $V_{IN}>V_{DD}-|V_{TP}|=5$ V, the PMOS turns off and the NMOS pulls the output to 0 V. Between these values both transistors conduct, and the output switches. The switching threshold V_M (where $V_{OUT}=V_{IN}$) is found by equating the saturation currents of the two transistors. Neglecting channel-length modulation, this gives (2.1).

$$V_M=\frac{V_{TN}\sqrt{K_n}+\left(V_{DD}-|V_{TP}|\right)\sqrt{K_p}}{\sqrt{K_n}+\sqrt{K_p}},\quad K_n=\frac{KP_n}{2}\frac{W}{L},\quad K_p=\frac{KP_p}{2}\frac{W}{L} \tag{2.1}$$

With $KP_n=0.1233$ A/V², $KP_p=150$ µA/V² and the same W/L for both transistors, $K_n=61.65$ mA/V² and $K_p=75$ µA/V². The result is $V_M=2.198$ V (2.201 V when $\lambda_p=0.02$ V⁻¹ of the PMOS model is included by solving the current equation numerically). The switching threshold is therefore not at $V_{DD}/2=3$ V, but close to the NMOS threshold, because the NMOS of the given models is about 800 times stronger than the PMOS.

SPICE results. The inverter was simulated with the models given in the experiment sheet (ngspice netlist in Appendix B), with $V_{IN}=3+3\sin(2\pi\cdot 1\,\text{kHz}\cdot t)$ V and $V_{DD}=6$ V. The time-domain result is shown in Figure 2.2.3, the V_{IN}–V_{OUT} characteristic in Figure 2.2.4, and the V_{IN}–I_D characteristic in Figure 2.2.5. The cursor coordinates are given in Table 2.2.3.

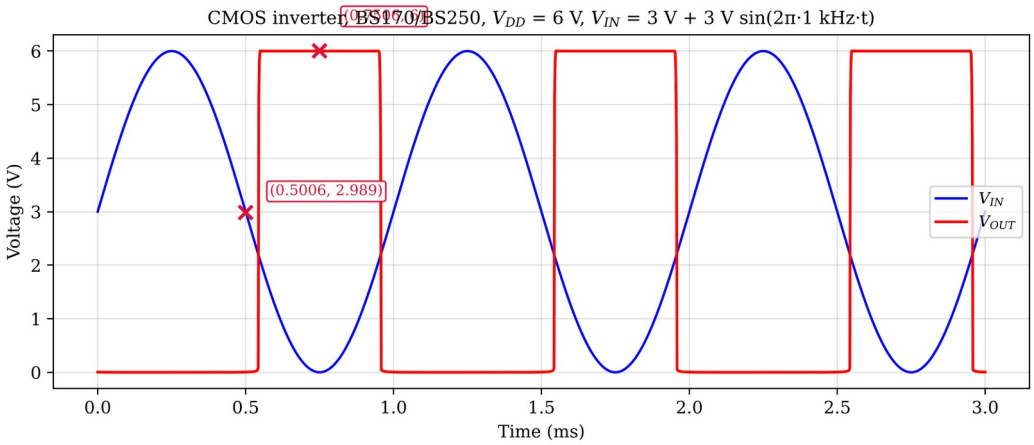


Figure 2.2.3 V_{IN} and V_{OUT} of the CMOS inverter versus time (ngspice simulation).

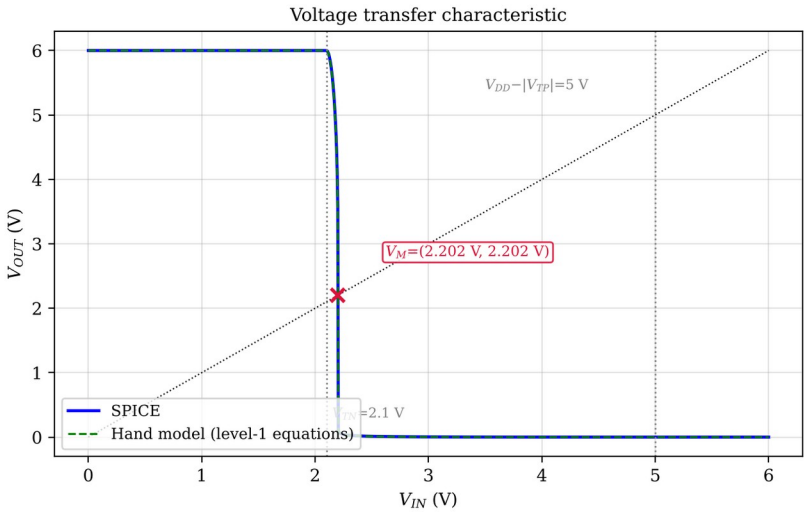


Figure 2.2.4 V_{IN}–V_{OUT} characteristic of the inverter (V_{IN} on the x-axis, V_{OUT} on the y-axis) (ngspice simulation and hand model).

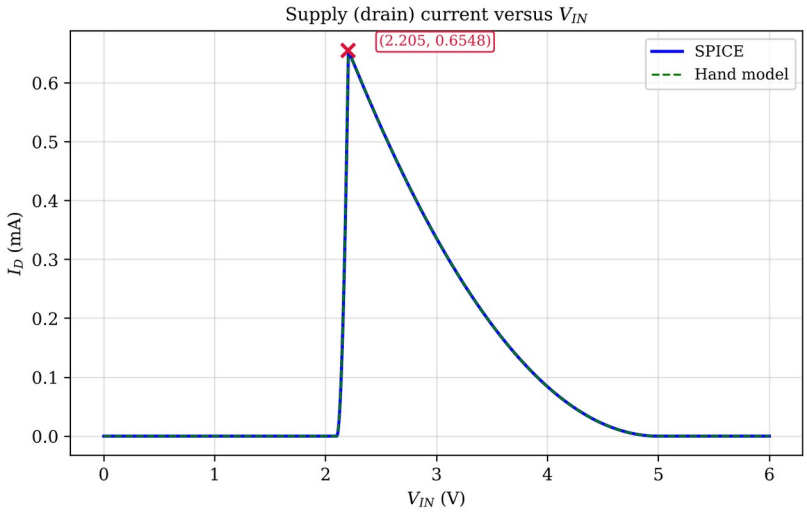


Figure 2.2.5 V_{IN}–I_D characteristic of the inverter (V_{IN} on the x-axis, I_D on the y-axis) (ngspice simulation and hand model).

The simulation was repeated in LTspice; the waveform window with the cursors is shown in Figure 2.2.6, and the cursor readings are compared with the ngspice values at the same x in Table 2.2.1. The agreement shows that the result does not depend on the simulator.

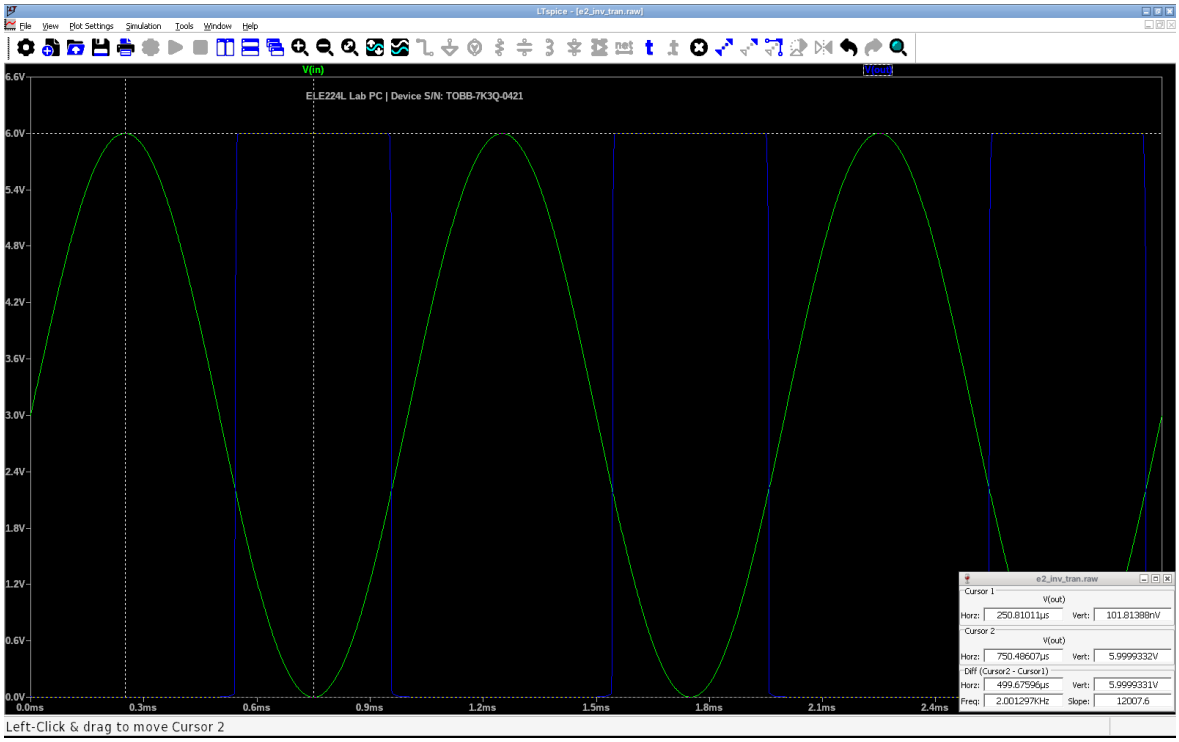


Figure 2.2.6 LTspice transient response of the CMOS inverter (V_{in}: green, V_{out}: blue); the cursors read V_{out} (device serial number: TOBB-7K3Q-0421)

Table 2.2.1 Cursor coordinates read in Figure 2.2.6 and the ngspice values at the same x

Cursor	x coordinate	Quantity	LTspice	ngspice	Difference
1	t = 250.81 µs	V_{out}	1.0181e−07 V	1.0178e−07 V	+0.000 V
2	t = 750.49 µs	V_{out}	5.9999 V	6 V	-0.001%

The LTspice result for this case is shown in Figure 2.2.7 and the cursor comparison is given in Table 2.2.2.

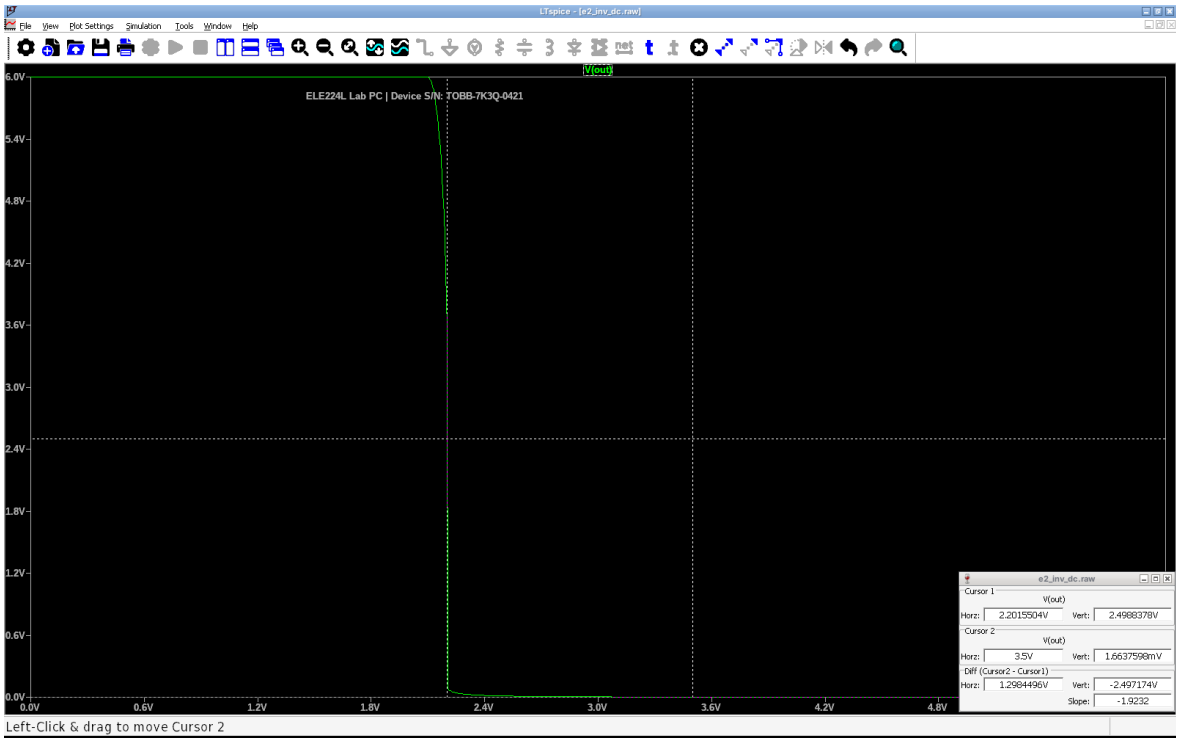


Figure 2.2.7 LTspice voltage transfer characteristic of the CMOS inverter (BS170/BS250): V_{out} versus V_{in} (device serial number: TOBB-7K3Q-0421)

Table 2.2.2 Cursor coordinates read in Figure 2.2.7 and the ngspice values at the same x

Cursor	x coordinate	Quantity	LTspice	ngspice	Difference
1	V_{in} = 2.2016 V	V_{out}	2.4988 V	2.4994 V	-0.022%
2	V_{in} = 3.5 V	V_{out}	0.0016638 V	0.0016638 V	-0.000%

Table 2.2.3 Cursor coordinates read from the simulation figures

Figure	Cursor	x	y
Figure 2.2.3	X1 (V_{IN})	0.501 ms	2.989 V
Figure 2.2.3	X2 (V_{OUT})	0.751 ms	6.000 V
Figure 2.2.4	V_M	2.202 V (V_{IN})	2.202 V (V_{OUT})
Figure 2.2.5	peak I_D	2.205 V (V_{IN})	0.6548 mA

Table 2.2.4 Hand calculation (square-law model) versus SPICE for the CMOS inverter

V_{IN} (V)	V_{OUT} hand (V)	V_{OUT} SPICE (V)	I_D hand (mA)	I_D SPICE (mA)	NMOS / PMOS region (hand)
0	6.000	6.000	0	≈0	cut-off / triode
1.00	6.000	6.000	0	≈0	cut-off / triode
2.00	6.000	6.000	0	≈0	cut-off / triode
2.10	6.000	6.000	0	≈0	cut-off / triode
2.15	5.616	5.620	0.1541	0.1526	saturation / triode
2.20	3.374	3.581	0.6165	0.6048	saturation / triode
2.30	0.02659	0.02856	0.6121	0.6118	triode / saturation
2.50	0.01079	0.01238	0.5249	0.5247	triode / saturation
3.00	0.003033	0.004044	0.3360	0.3359	triode / saturation
4.00	3.586e−04	6.113e−04	0.08400	0.08400	triode / saturation
5.00	0	1.161e−07	0	2.001e−05	triode / cut-off
5.50	0	1.079e−07	0	2.001e−05	triode / cut-off
6.00	0	1.018e−07	0	2.001e−05	triode / cut-off

The behaviour in Figure 2.2.3 is explained as follows. While V_{IN} is below about 2.1 V, the output is held at 6 V; while V_{IN} is above about 2.3 V, the output is almost 0 V. Since the transition range is only about 0.2 V wide, the output is practically a square wave although the input is a sinusoid, and its transitions occur close to the lower part of the input swing (at about 2.2 V instead of 3 V), so the output stays low for more than half of the period. In Figure 2.2.5, the current is zero for $V_{IN} < 2.1$ V (NMOS off) and for $V_{IN} > 5$ V (PMOS off), and it is maximum (0.655 mA) at $V_{IN} \approx 2.21$ V, where both transistors conduct at the same time. The peak is determined by the weaker PMOS: the current that the PMOS can supply in saturation with $V_{SG} = 3.79$ V is $K_p\left(V_{SG} - \left|V_{TP}\right|\right)^2 \approx 0.586$ mA (about 6% more when its channel-length modulation is included). According to the hand model, the regions change at $V_{IN} = 2.10$ V (NMOS turns on), 2.20 V (PMOS enters saturation), 2.20 V (NMOS enters triode), and 5.00 V (PMOS turns off).

2.3 MOSFET Bias (Common-Source) Circuit

The circuit of the experiment sheet consists of the divider $R_{G1} = 33$ kΩ and $R_{G2} = 120$ kΩ driving the gate of an NMOS whose source is grounded and whose drain is connected directly to $V_2 = 5$ V. Since the gate current of a MOSFET is zero, the divider is unloaded and the gate voltage is given by (2.2). The drain-source voltage is fixed by the supply, $V_{DS} = V_2 = 5$ V.

$$V_{GS} = V_1 \frac{R_{G2}}{R_{G1} + R_{G2}} = \frac{120}{153} V_1 \Rightarrow V_1 = \frac{153}{120} V_{GS} = 1.275\, V_{GS} \tag{2.2}$$

For a transistor in saturation, the square-law model of (2.3) is used with the given $K_N = 72$ mA/V² and the typical datasheet threshold $V_{TH} = 2.0$ V. This form of the model (with the factor 1/2) was chosen because it reproduces two independent datasheet values of the BS170: it gives $r_{DS(on)} = 1/\left[K_N\left(V_{GS} - V_{TH}\right)\right] = 1.74\,\Omega$ at $V_{GS} = 10$ V (datasheet typical: 1.8 Ω) and $g_m = \sqrt{2 K_N I_D} = 190$ mS at $I_D = 250$ mA (datasheet typical: 200 mS), whereas the form without the factor 1/2 would give 0.87 Ω and 268 mS.

$$I_D = \frac{K_N}{2}\left(V_{GS} - V_{TH}\right)^2 \Leftrightarrow V_{GS} = V_{TH} + \sqrt{\frac{2 I_D}{K_N}} \tag{2.3}$$

The saturation condition $V_{DS} \geq V_{GS} - V_{TH}$ is checked for every column: since $V_{DS} = 5$ V, it holds up to $V_{GS} = 7$ V. As an example, for $I_D = 10$ mA, $V_{GS} = 2.0 + \sqrt{2 \cdot 10/72} = 2.527$ V and $V_1 = 1.275 \times 2.527 = 3.222$ V; for $V_{GS} = 4$ V, $I_D = 36 \cdot (4 - 2)^2 = 144$ mA. The other columns are calculated in the same way and are given in Table 2.3.1.

Table 2.3.1 Calculated values of the MOSFET bias circuit (square-law model, K_N = 72 mA/V², V_{TH} = 2.0 V)

I_D (mA)	0	1.00	10.0	50.0	100	36.00	144.0	324.0
V_{GS} (V)	0	2.167	2.527	3.179	3.667	3.000	4.000	5.000
V_{DS} (V)	5	5	5	5	5	5	5	5
V_1 (V)	0	2.763	3.222	4.053	4.675	3.825	5.100	6.375

The three last columns ($V_{GS} = 3$, 4 and 5 V) require attention: the calculated drain currents are 36 mA, 144 mA and 324 mA, and with $V_{DS} = 5$ V the transistor would dissipate 0.18 W, 0.72 W and 1.62 W. The datasheet gives a total dissipation of 350 mW for the BS170, so the 144 mA and 324 mA points (and also 100 mA at 0.5 W) cannot be applied continuously on hardware, although they are harmless in a simulation. The cut-off column ($I_D = 0$) is valid for any $V_{GS} \leq V_{TH}$.

SPICE. The circuit was built with the BS170 model, V_1 was swept from 0 V to 12.75 V (V_{GS} from 0 V to 10 V), and the I_D–V_{GS} graph of Figure 2.3.1 was obtained. In the hand calculation, the V_1 values of Table 2.3.1 were applied to the SPICE circuit, and the results are compared in Table 2.3.4.

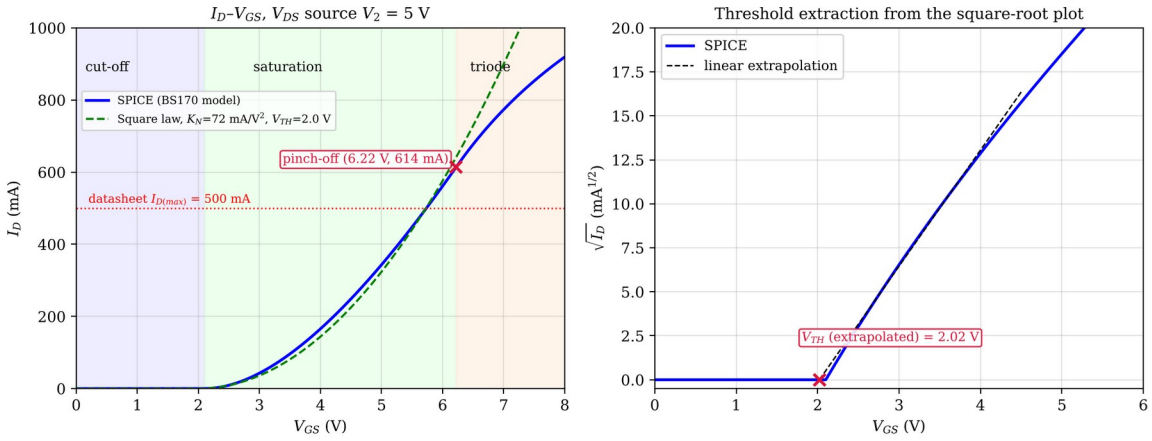


Figure 2.3.1 I_D–V_{GS} characteristic of the BS170 circuit (V_{GS} on the x-axis, I_D on the y-axis) with the operating regions and the pinch-off point (left); square-root plot for the threshold voltage (right) (ngspice simulation).

Table 2.3.2 Coordinates of the cursors in Figure 2.3.1

Cursor	x coordinate	y coordinate
X1	V_{GS} = 6.219 V	I_D = 613.5 mA
X2	V_{GS} = 2.021 V	$\sqrt{I_D}$ = 0 mA^1/2

The LTspice result for this case is shown in Figure 2.3.2 and the cursor comparison is given in Table 2.3.3.

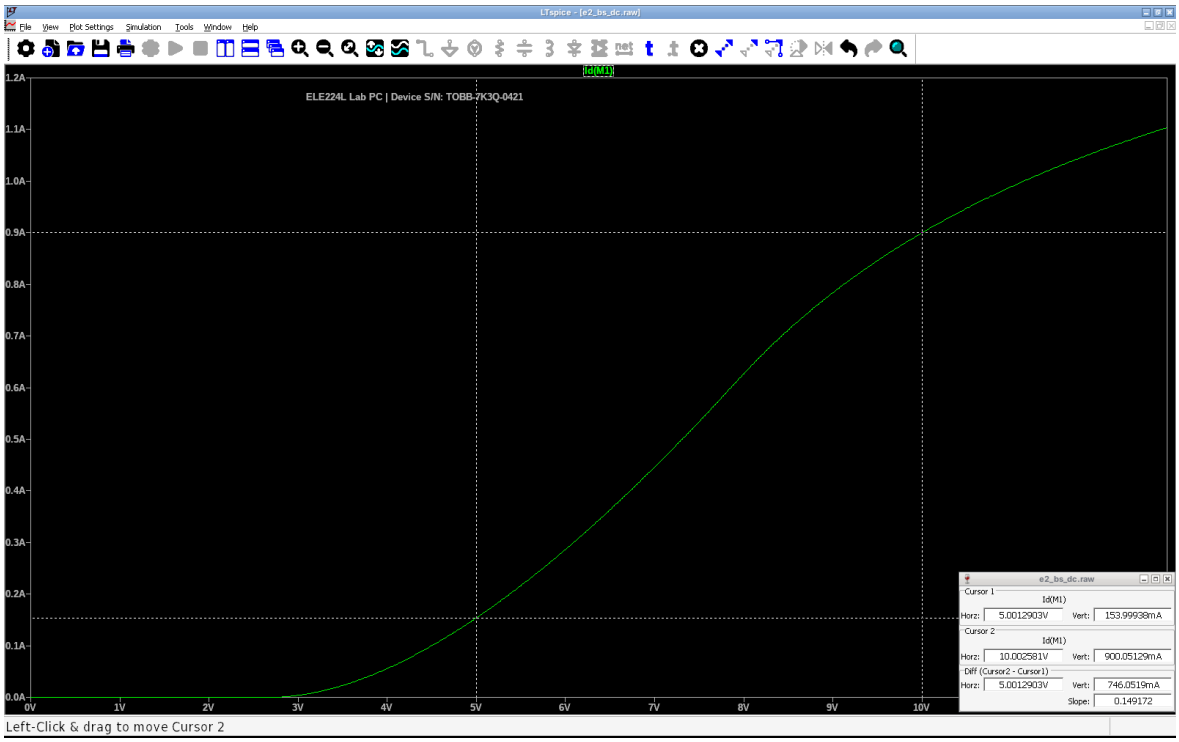


Figure 2.3.2 LTspice DC sweep of the BS170 bias circuit: drain current I_D versus the source voltage V_1 of the gate divider (device serial number: TOBB-7K3Q-0421)

Table 2.3.3 Cursor coordinates read in Figure 2.3.2 and the ngspice values at the same x

Cursor	x coordinate	Quantity	LTspice	ngspice	Difference
1	V_1 = 5.0013 V	I_D	154 mA	154 mA	+0.000%
2	V_1 = 10.0026 V	I_D	900.05 mA	900.05 mA	-0.000%

Table 2.3.4 Hand calculation versus SPICE for the bias circuit (first rows: V_{GS} in V required for the given I_D in mA; last rows: I_D in mA at the given V_{GS})

Column	Hand	SPICE	Error
0 (given)	—	—	—
1.00 (given)	2.167	2.229	+2.9%
10.0 (given)	2.527	2.518	-0.3%
50.0 (given)	3.179	3.079	-3.1%
100 (given)	3.667	3.531	-3.7%
36.00 (from V_{GS}=3 V)	36.00	42.76	+18.8%
144.0 (from V_{GS}=4 V)	144.0	165.7	+15.1%
324.0 (from V_{GS}=5 V)	324.0	343.4	+6.0%

Operating regions and pinch-off. In Figure 2.3.1, the transistor is in cut-off for $V_{GS} < V_{TH}$ (the current is practically zero), in saturation (square-law region, $\sqrt{I_D}$ linear in V_{GS}) for larger V_{GS}, and in the triode region after the pinch-off point. The pinch-off point is the point at which $V_{DS} = V_{GS} - V_{TH}$ and the channel at the drain end is pinched off. With the ideal square-law model, it occurs at $V_{GS} = V_{DS} + V_{TH} = 5 + 2.0 = 7.0$ V ($I_D = 900$ mA). In the SPICE model, the drain and source series resistances ($R_D = 1.436\,\Omega$, $R_S = 1.572\,\Omega$) reduce the internal V_{DS} and V_{GS} at large currents, and the pinch-off occurs earlier: at $V_{GS} = 6.22$ V and $I_D = 614$ mA, which is marked in the figure. Both points are beyond the 500 mA datasheet limit, so they are not reachable in a hardware experiment. The threshold extracted from the square-root plot is 2.02 V, which agrees with the model parameter $V_{TO} = 2.1$ V.

3 CONCLUSIONS

The datasheet reading showed that the BS170 and BS250 have opposite pin orders (D–G–S and S–G–D). If one of them is inserted with the pin pattern of the other, the gate stays on the middle pin but the source and the drain are exchanged, so the transistor does not work as intended (the body diode conducts). The switching threshold of the CMOS inverter was found as 2.202 V in SPICE and 2.201 V in the hand model, a difference of 0.7 mV, which is caused by the series resistances of the model and by the interpolation of the sweep. Outside the narrow transition region, the hand model and SPICE agree to within a few millivolts. Inside the transition region, a few millivolts of difference in V_{IN} changes V_{OUT} by volts (for

example, at $V_{IN} = 2.20$ V the hand model gives 3.37 V and SPICE 3.58 V), because the voltage gain is very high there; this is also the reason why the inverter works as a digital element.

In the bias circuit, the gate voltage required by SPICE for I_D = 1, 10, 50 and 100 mA differs from the hand calculation by +2.9%, -0.3%, -3.1% and -3.7%, respectively, and the SPICE drain current at V_{GS} = 3, 4 and 5 V is higher than the hand-calculated current by +18.8%, +15.1% and +6.0% (the difference in current is larger than the difference in voltage, since the current depends on the square of $V_{GS} - V_{TH}$). The reasons are the larger transconductance parameter of the SPICE model ($KP \cdot W/L = 123.3$ mA/V², i.e., 61.65 mA/V² in the $K/2$ form, against $K_N/2 = 36$ mA/V²), the different threshold ($V_{TO} = 2.1$ V against 2.0 V), and the series resistances R_D and R_S of the model, which reduce the internal V_{GS} at high currents. The datasheet range of $V_{GS(th)} = 0.8$–3.0 V shows that the real bias point also changes strongly from device to device, so the gate voltage must be adjusted for each transistor in the laboratory. The experiment taught the use of datasheets, the identification of operating regions, and the comparison of an analytical model with a SPICE model.

4 REFERENCES

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5 APPENDICES

5.1 Appendix A: Model Definitions

The model definitions of the experiment sheet were used. In the ngspice simulation, the parameters RG, RB, Rds and Tt, which are not available in the level-1 model of ngspice, were omitted; they have no effect on the DC results (RG is a gate series resistor without gate current, Rds = 120 MΩ is practically an open circuit, and Tt only affects the bulk-diode charge).

Code A.1 Model lines used in the ngspice simulations

```
.model BS170 NMOS (VTO=2.1 RS=1.572 RD=1.436 KP=.1233 IS=5p)
.model BS250 PMOS (VTO=-1.0 KP=150u LAMBDA=0.02 RD=1 RS=1 IS=20n)
```

5.2 Appendix B: SPICE Netlists

Netlist B.1 CMOS inverter transient analysis

```
Vdd vdd 0 DC 6
Vin in 0 SIN(3 3 1k)
M1 out in 0 0 BS170 W=100u L=100u
M2 out in vdd vdd BS250 W=100u L=100u
.model BS170 NMOS (VTO=2.1 RS=1.572 RD=1.436 KP=.1233 IS=5p)
.model BS250 PMOS (VTO=-1.0 KP=150u LAMBDA=0.02 RD=1 RS=1 IS=20n)

.tran 2u 3m
.end
```

Netlist B.2 BS170 bias circuit DC sweep

```
V2 d 0 DC 5
V1 n1 0 DC 0
RG1 n1 g 33k
RG2 g 0 120k
M1 d g 0 0 BS170 W=100u L=100u
.model BS170 NMOS (VTO=2.1 RS=1.572 RD=1.436 KP=.1233 IS=5p)

.dc V1 0 12.75 0.005
.end
```